

Cellular Networks

Advanced Mobile Communication Networks



Integrated Communication Systems Group

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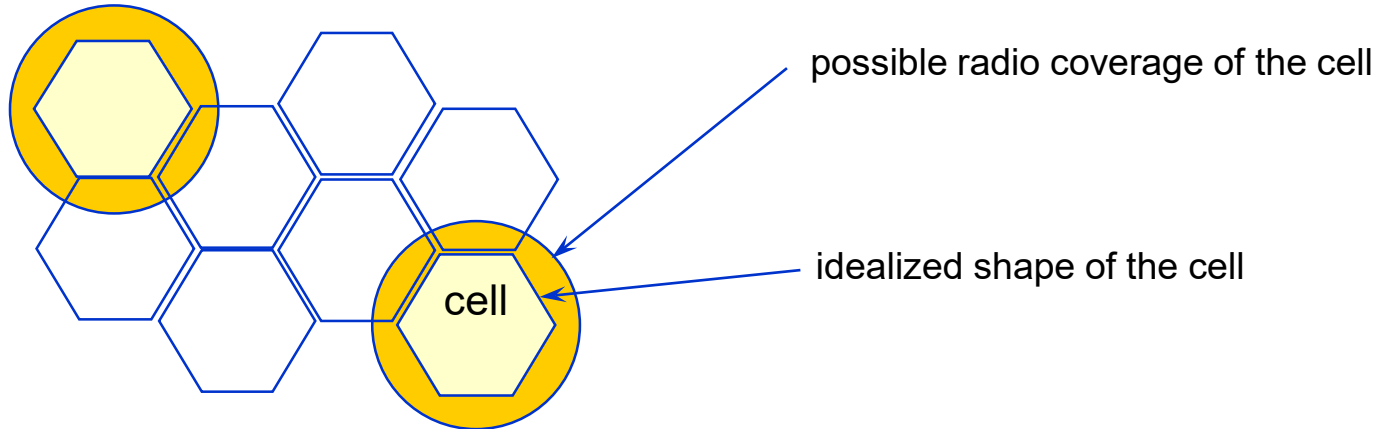
GSM, GPRS & EDGE

Questions to be answered:

- What is the idea behind GSM? What is its access scheme? Why is FDD preferred over TDD? Describe the mixture of FDMA and TDMA used for GSM.
- GSM handover: what's the idea? What's the difference to a handover in a WLAN system?
- Is there a handover if a GSM mobile does not have an active call? Why or why not?
- How does authentication work in GSM? Who or what is authenticated? How is the integrity of the shared key ensured?
- What is the difference between hard and soft handovers?
- What is the difference between make-after-break and make-before-break?
- What are the main differences between a GSM and a WLAN system? Compare the two systems with respect to media access, radio resource management and handover execution.
- Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate) for GSM voice and GPRS/EDGE.

GSM: cellular network

Segmentation of the area into cells



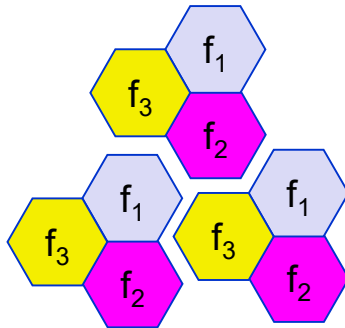
- use of several carrier frequencies
- different frequencies in neighbor cells
- cell radius varies from some 100 m up to 35 km depending on user density, geography, transceiver power etc.
- hexagonal shape of cells is idealized (cells overlap, shapes depend on geography)
- if a mobile user changes cells
 - > handover of the connection to the neighbor cell

Cellular systems: Frequency planning

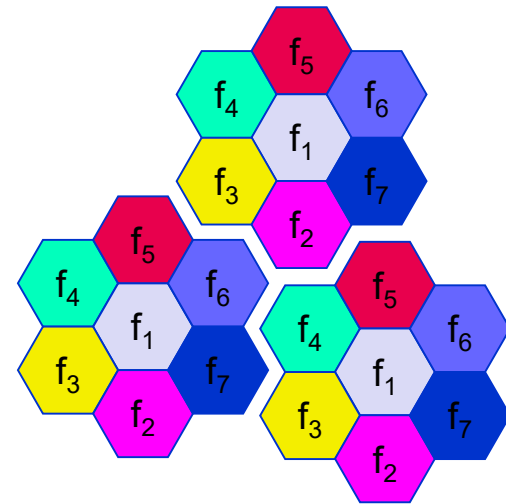
Frequency reuse only beyond a certain distance between base stations

Typical (hexagon) model:

reuse-3 cluster:



reuse-7 cluster:



Other regular pattern: reuse-19

→ Frequency reuse pattern determines the experienced SIR

Fixed frequency assignment:

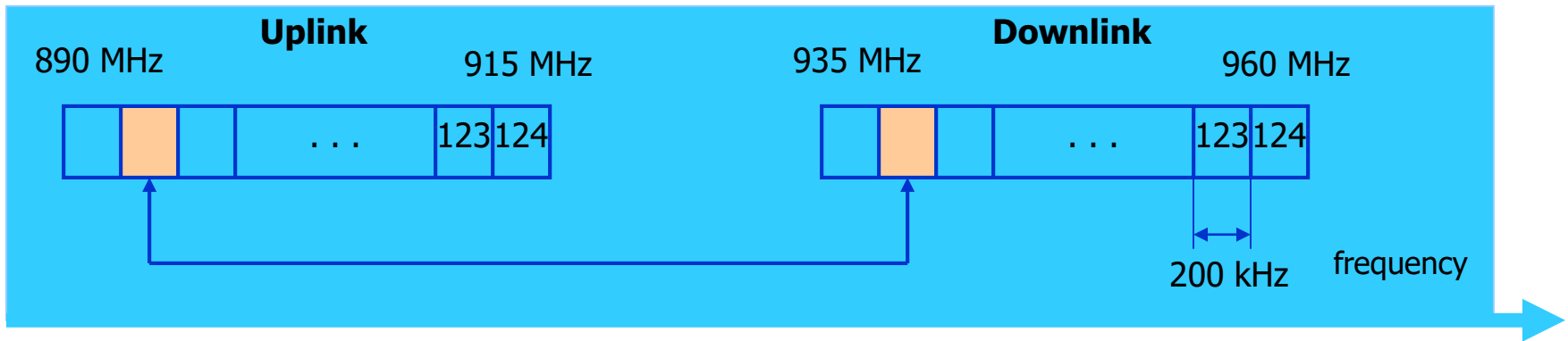
- certain frequencies are assigned to a certain cell
- problem: different traffic load in different cells

Frequency Hopping:

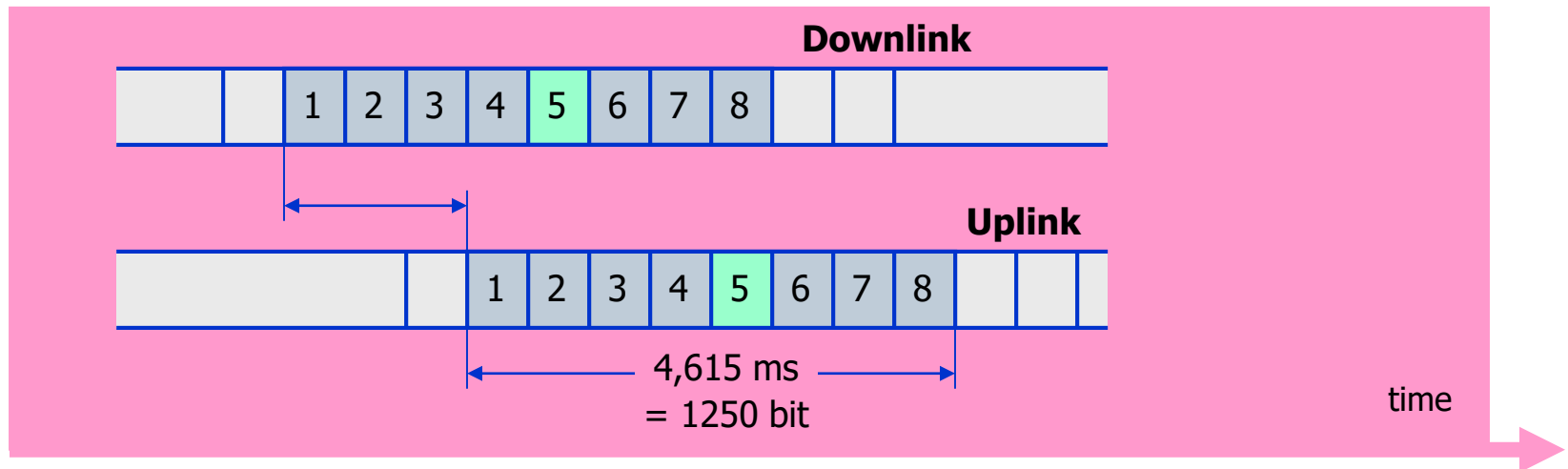
- Improves quality for slow moving or stationary users (frequency diversity)
- Reduces impact of intercell interference by statistical averaging

GSM: Air Interface

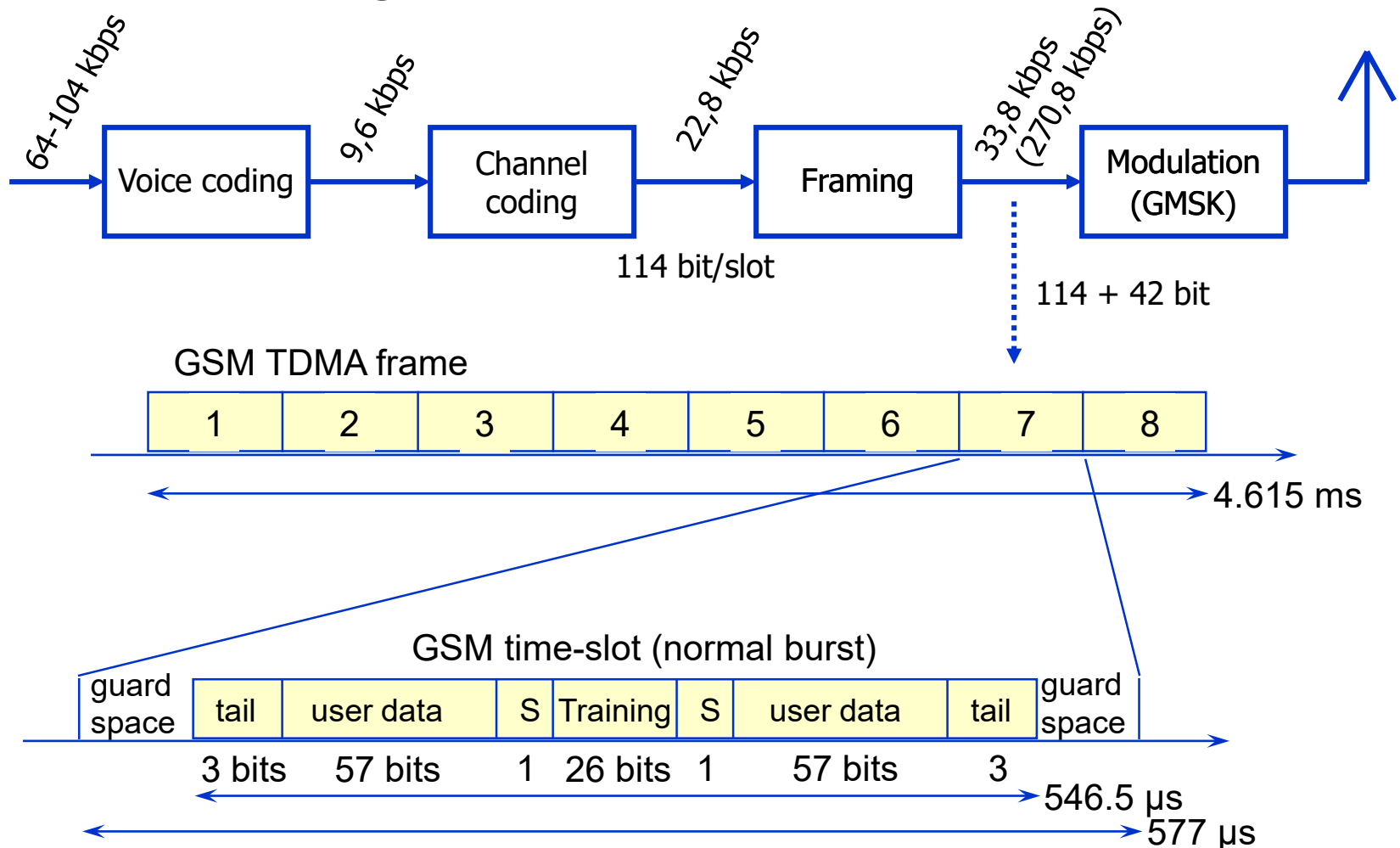
FDMA (Frequency Division Multiple Access) / **FDD** (Frequency Division Duplex)



TDMA (Time Division Multiple Access)



GSM: Voice Coding



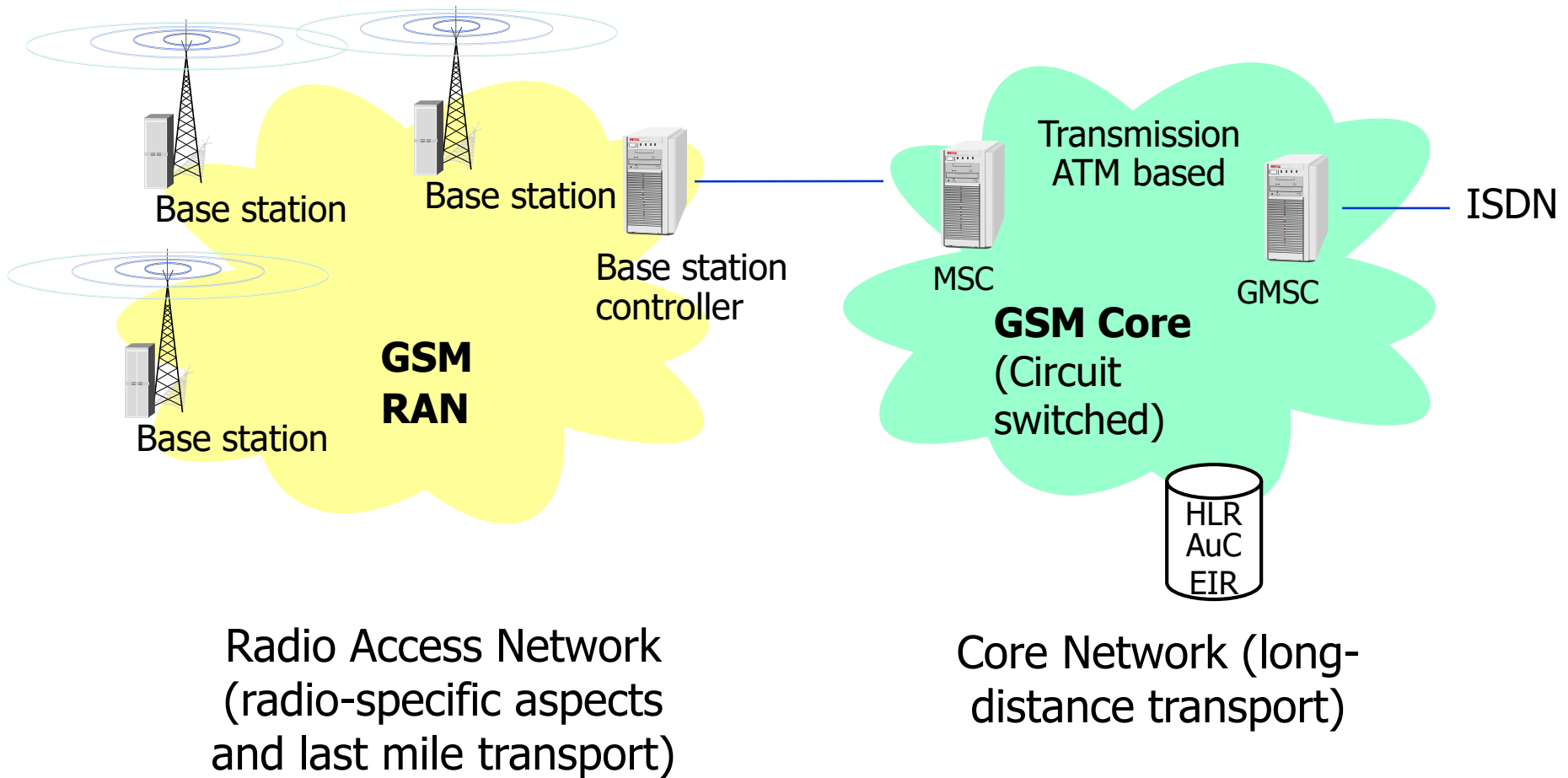
Guard (8.25 bits): avoid overlap with other time slots (different time offset of neighboring slot)

Training sequence: select the best radio path in the receiver and train equalizer

Tail: needed to enhance receiver performance

Flag S: indication for user data or control data

GSM Architecture



GSM system: Network elements

GSM is a **PLMN** (Public Land Mobile Network)

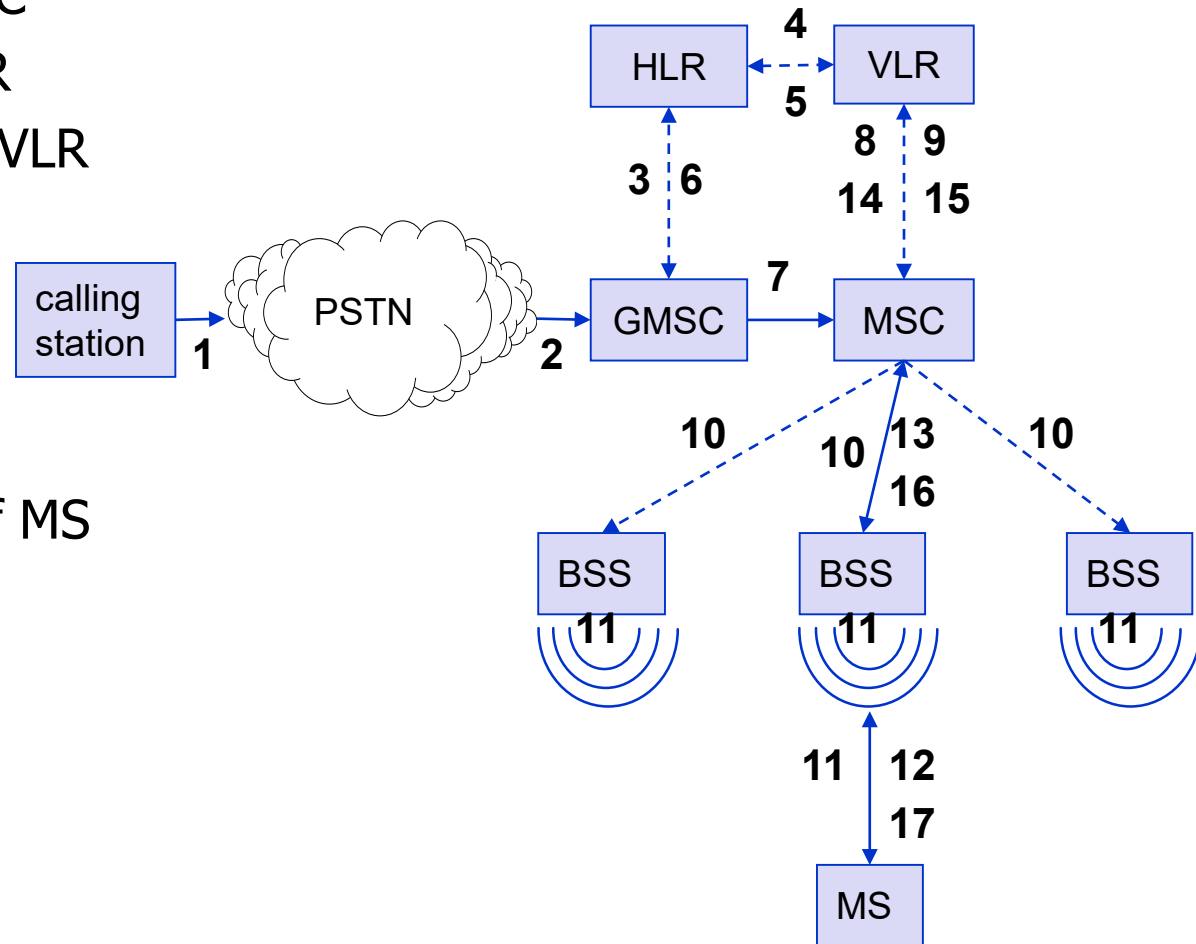
- several providers setup mobile networks following the GSM standard within each country

GSM Elements:

- MS (mobile station)
- Radio Access Network (Base station subsystem - BSS): covers all radio aspects
 - BTS (base transceiver station)
 - BSC (base station controller)
- Core Network: call forwarding, handover, switching
 - MSC (mobile services switching center)
 - GMSC (gateway MSC)
 - HLR (home location register) and VLR (visiting location register)
 - OMC (operation and maintenance center)
 - AuC (authentication center)
 - EIR (equipment identity register)

Mobile Terminated Call (MTC)

- 1: calling a GSM subscriber
- 2: forwarding call to GMSC
- 3: signal call setup to HLR
- 4, 5: request MSRN from VLR
- 6: forward responsible MSC to GMSC
- 7: forward call to current MSC
- 8, 9: get current status of MS
- 10, 11: paging of MS
- 12, 13: MS answers
- 14, 15: security checks
- 16, 17: set up connection



Location Management / Mobility Management

The issue: Compromise between

- minimizing the area where to search for a mobile
- minimizing the number of location updates

Solution 1:
Large paging area

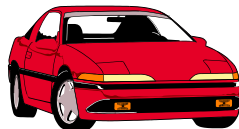
Location Area (LA)

Location Update

TOTAL
Signalling Cost =

$\sum$ Paging
Signalling Cost

+ $\sum$ Paging Area Update
Signalling Cost



Location Update

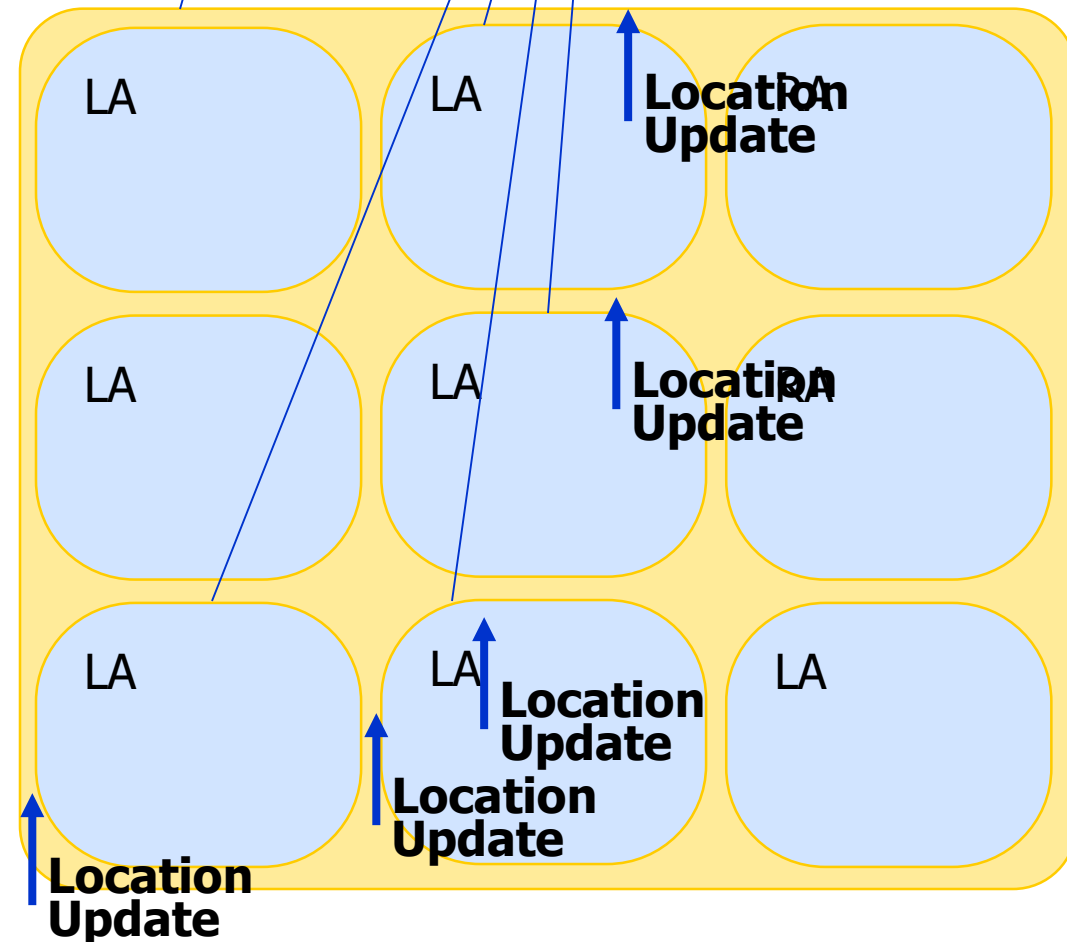
Location Management / Mobility Management

The issue: Compromise between

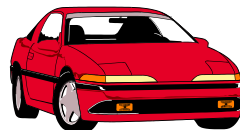
- minimizing the area where to search for a mobile
- minimizing the number of location updates

Solution 1:
Large paging area

Solution 2:
Small paging area



$$\begin{aligned} \text{TOTAL} \\ \text{Signalling Cost} &= \\ &\sum \text{Paging} \\ &\quad \text{Signalling Cost} \\ &+ \sum \text{Paging Area Update} \\ &\quad \text{Signalling Cost} \end{aligned}$$



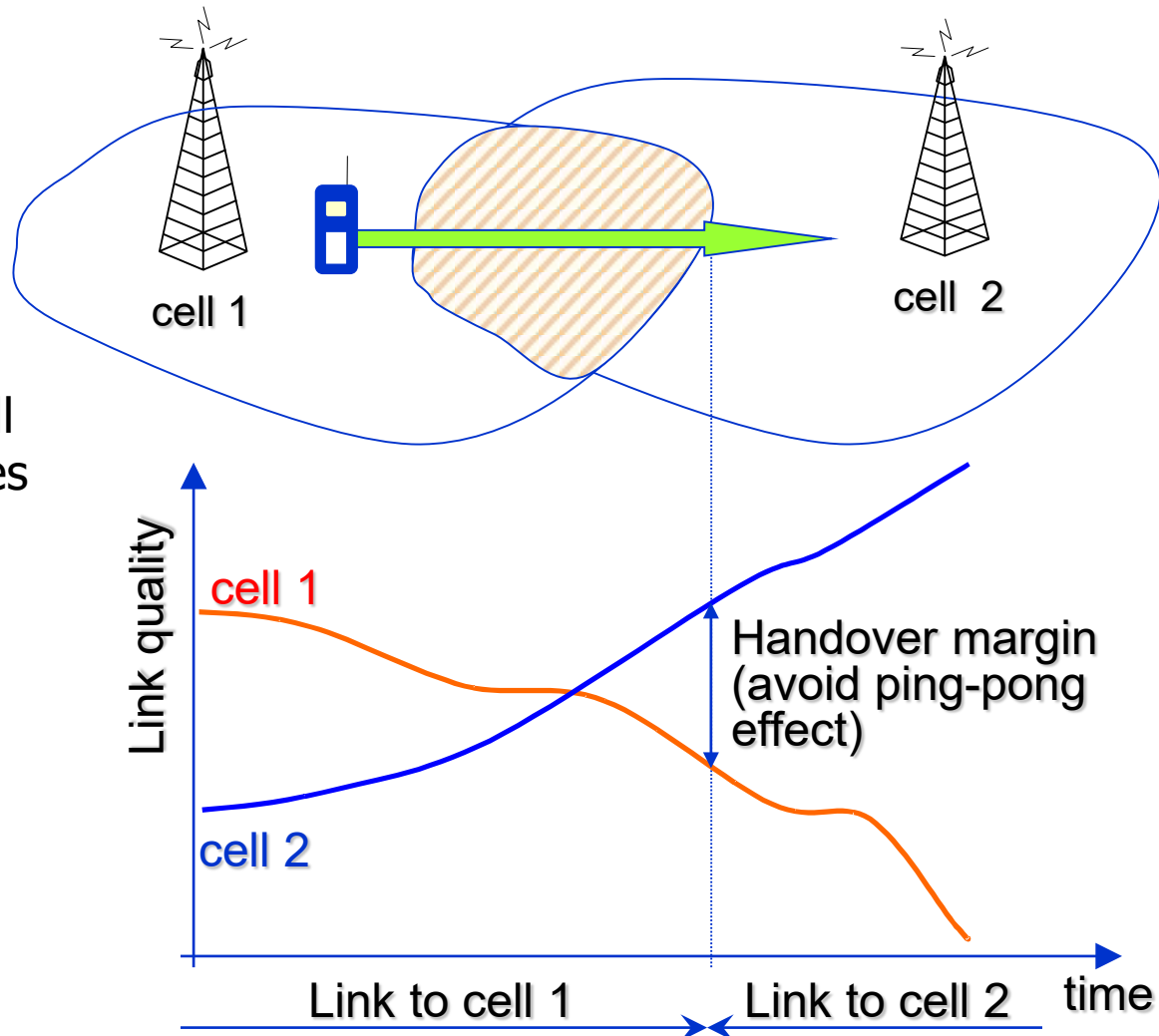
Handover

The problem:

Change the cell while communicating

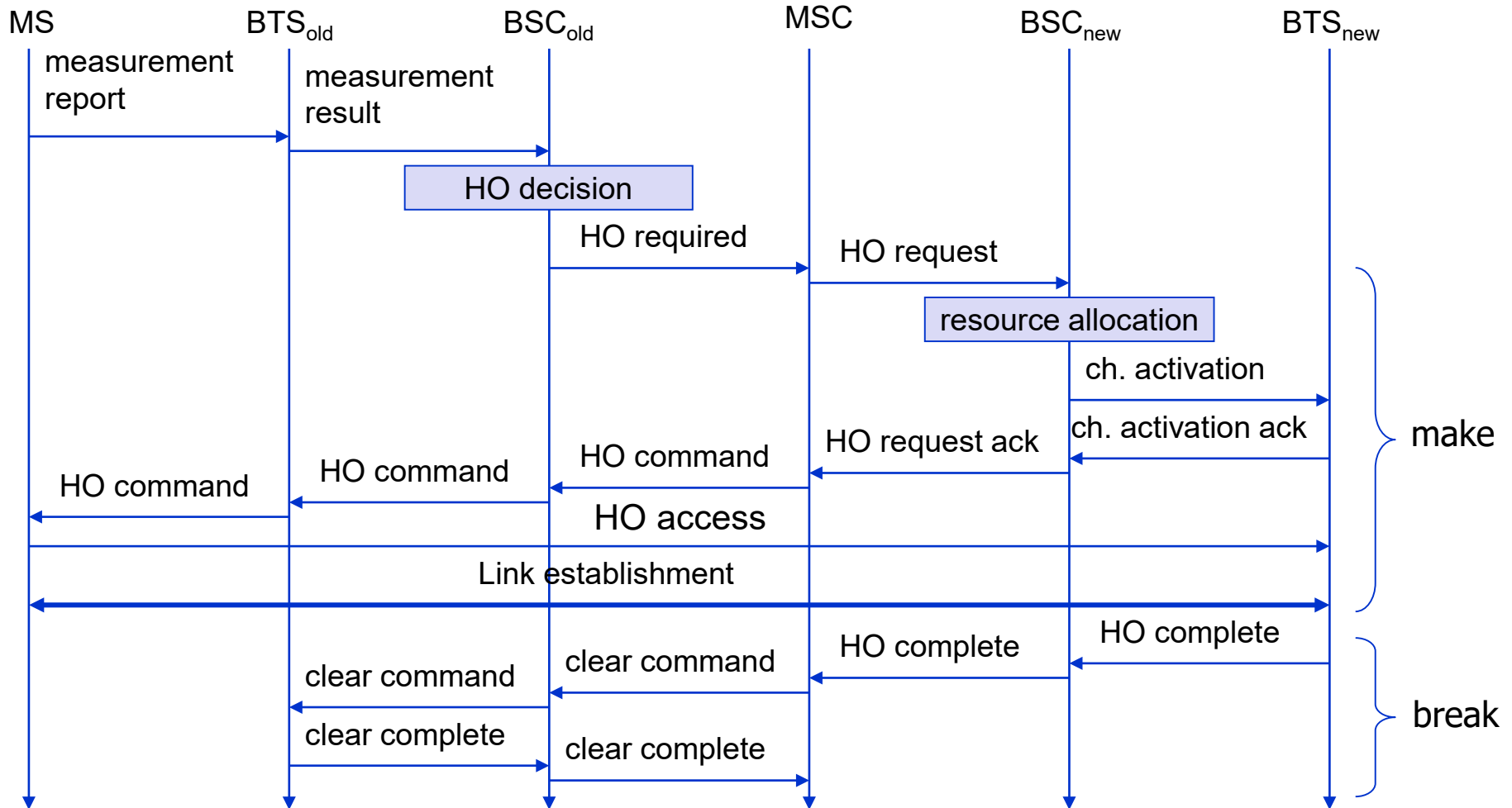
Reasons for handover:

- Quality of radio link deteriorates
- Communication in other cell requires less radio resources
- Supported radius is exceeded (e.g. Timing advance in GSM)
- Overload in current cell
- Maintenance



Handover procedure (change of BSC)

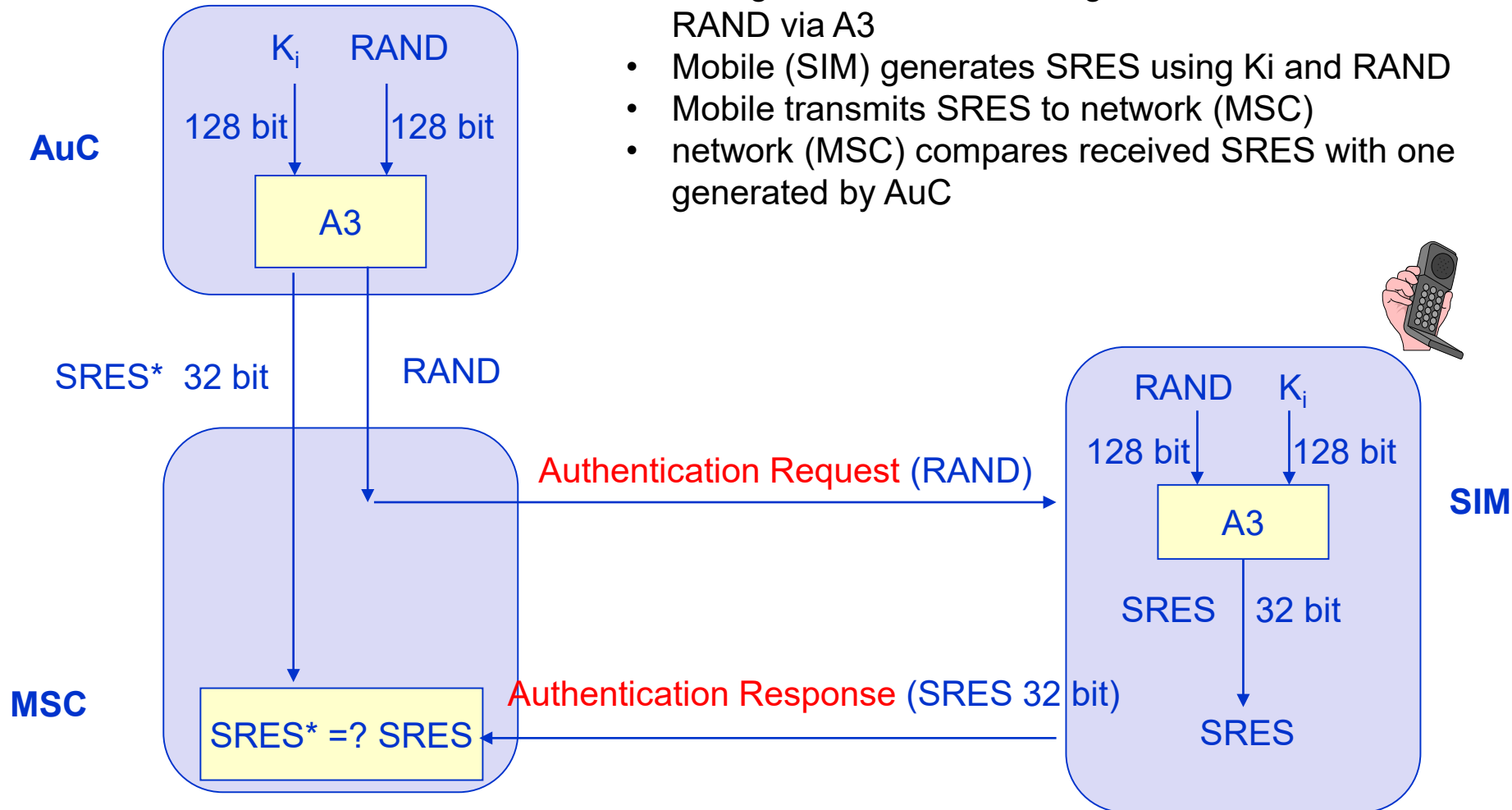
„Make-before-break“ strategy



GSM - authentication

Challenge-Response:

- Authentication center provides RAND to Mobile
- AuC generates SRES using K_i of subscriber and RAND via A3
- Mobile (SIM) generates SRES using K_i and RAND
- Mobile transmits SRES to network (MSC)
- network (MSC) compares received SRES with one generated by AuC



K_i : individual subscriber authentication key

SRES: signed response

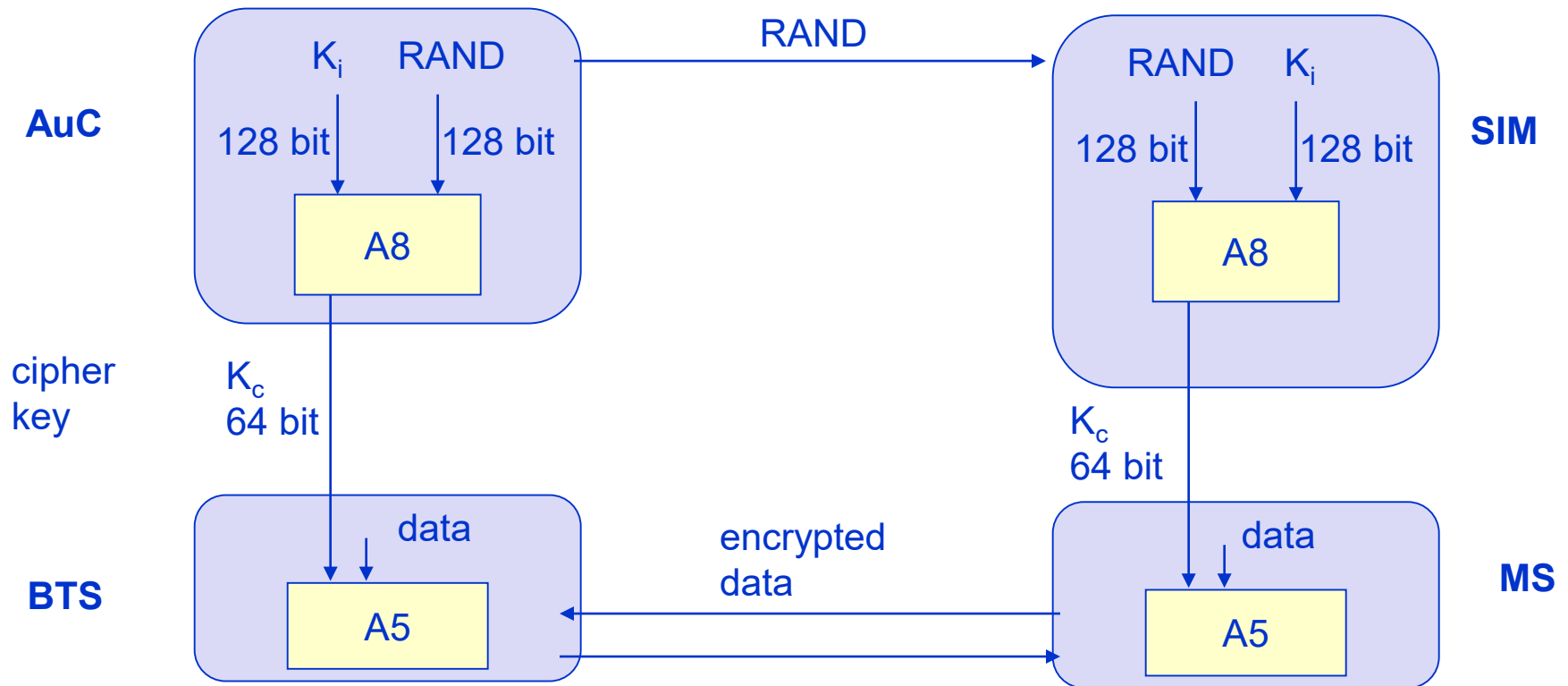
GSM - key generation and encryption

Ciphering:

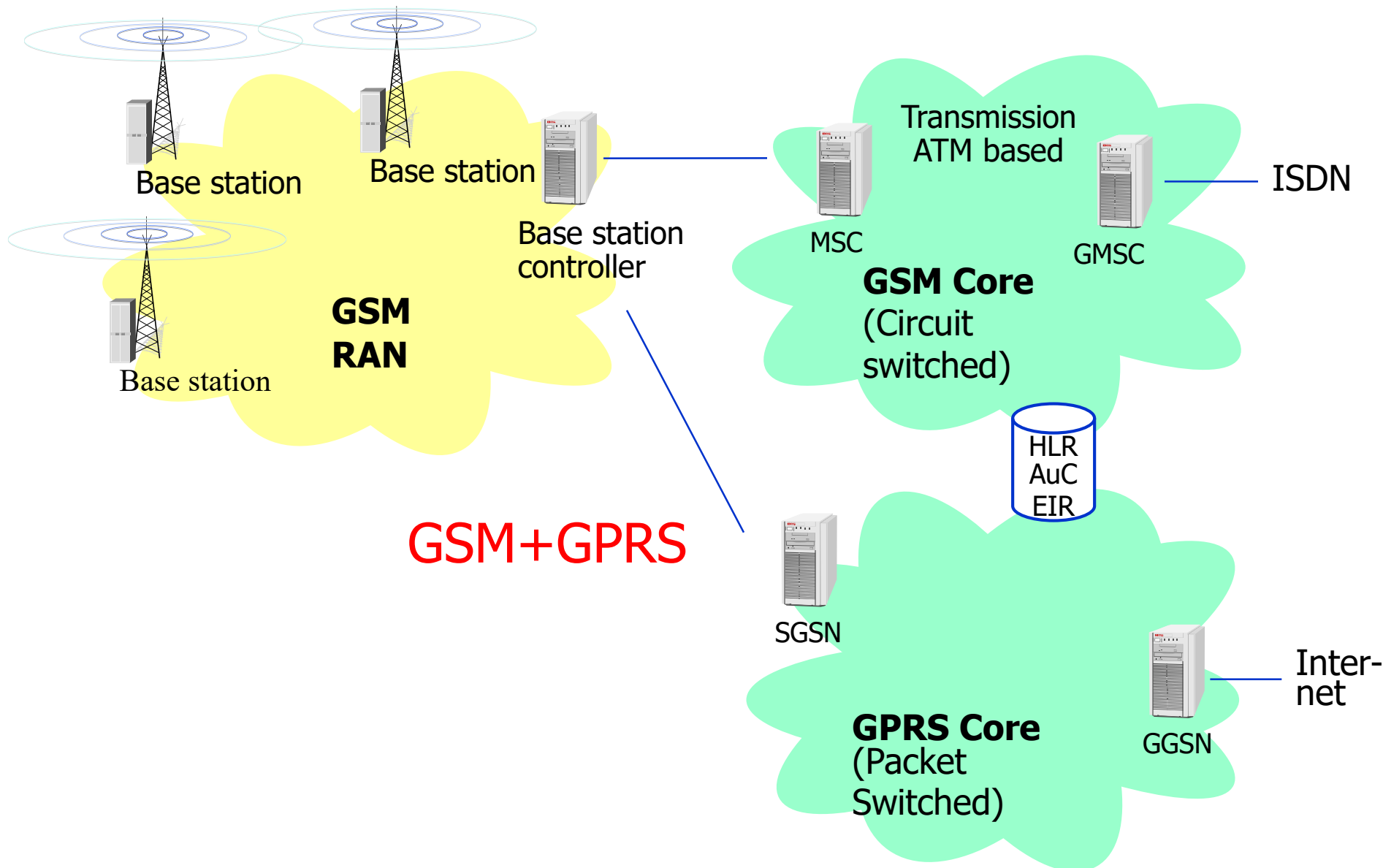
- Data sent on air interface ciphered for security
- A8 algorithm used to generate cipher key
- A5 algorithm used to cipher/decipher data
- Ciphering Key is never transmitted on air



MS with SIM



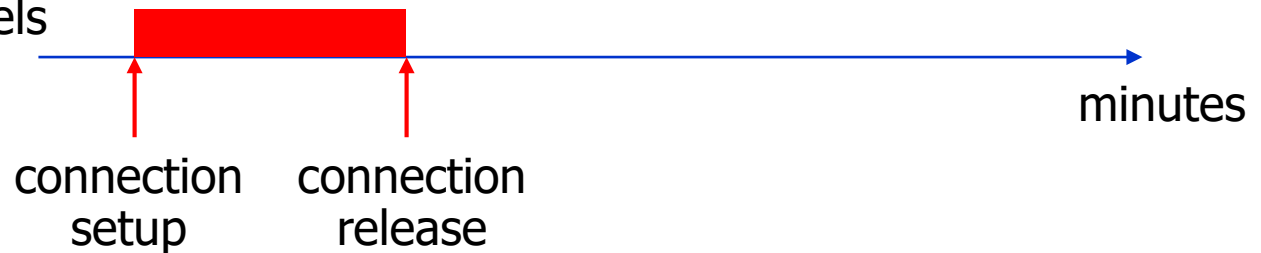
2G to 3G Evolution: GSM - GPRS - UMTS



Circuit vs. Packet Switched Communication

Connection (e.g. voice, CS data) => principle for GSM & UTRAN design

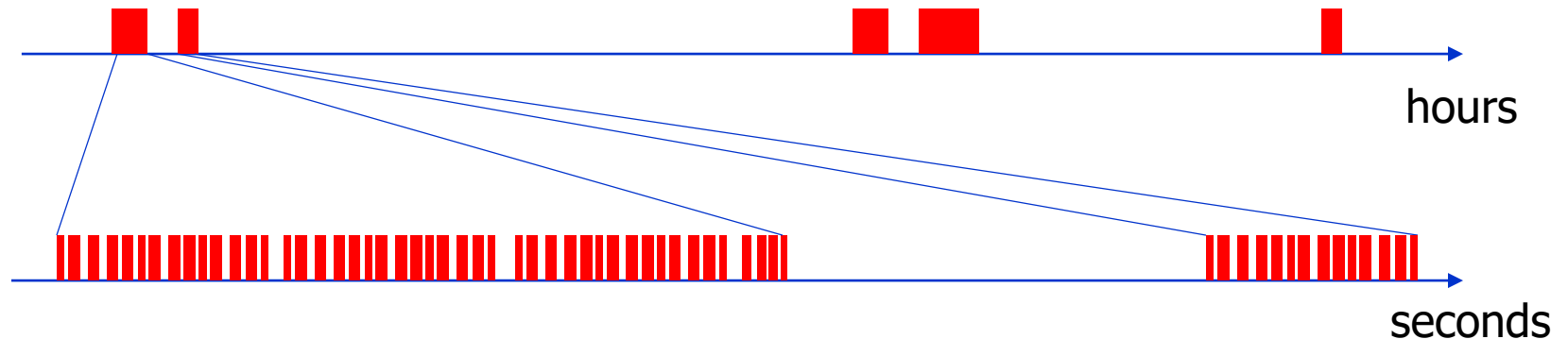
- clearly defined start and end times
- no burstiness
- => dedicated channels



Packet session => supported by GPRS/UMTS core, HSDPA, LTE

- packet arrival times are typically unknown to the system
- traffic is highly bursty

=> shared channels & packet scheduling



GPRS (General Packet Radio Service)

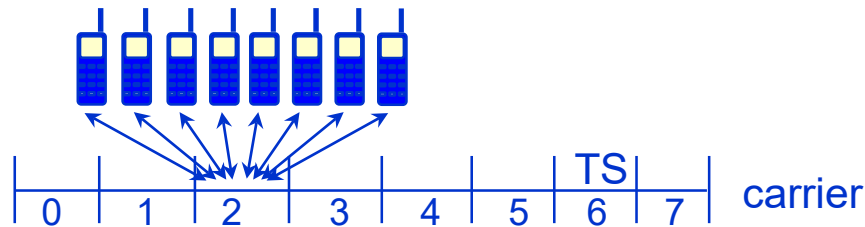
- Introducing packet switching in the network
- Using shared radio channels for packet transmission over the air:
 - multiplexing multiple MS on one time slot
 - flexible (also multiple) allocation of timeslots to MS (scheduling by PCU Packet Control Unit in BSC or BTS)
- using free slots only if data packets are ready to send (e.g., 115 kbit/s using 8 slots temporarily)
- standardization 1998, introduction 2001
- advantage: first step towards UMTS, flexible data services

GPRS network elements

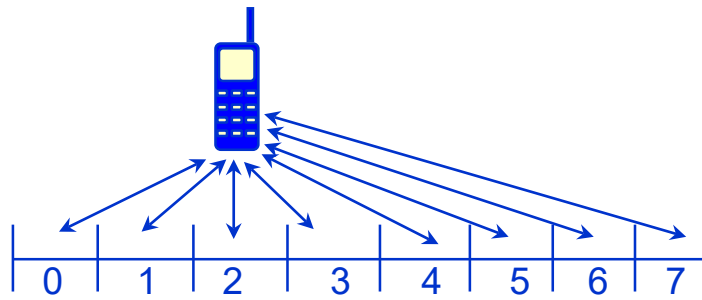
- GSN (GPRS Support Nodes): GGSN and SGSN
- **GGSN** (Gateway GSN)
 - interworking unit between GPRS and PDN (Packet Data Network)
- **SGSN** (Serving GSN)
 - supports the MS (location, billing, security)
- **HLR** (GPRS Register – GR)
 - maintains location and security information

GPRS: Multiplexing and multislots allocation

Multiplexing



Multislots capability



GPRS services

End-to-end packet switched traffic (peak channel rates)

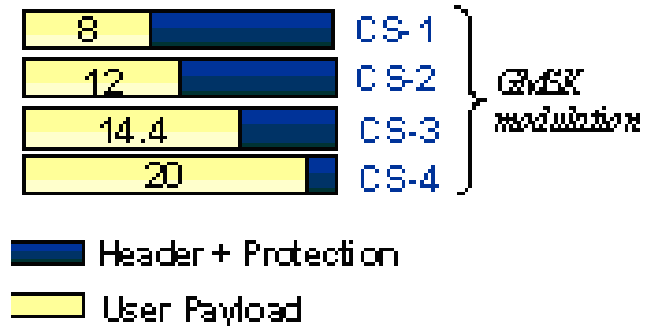
- 28 kbps (full use of 3 time slots, CS-1: FEC)
- 171.2 kbps (full use of 8 time slots, CS-4: no FEC)

Average aggregate throughput of a cell (Source: H. Menkes, WirelessWeb, Aug. 2002)

- 95 kbps (for both up and downlink)
- Assumptions: 4/12 reuse, realistic RF conditions, random traffic
- Worse figures for individual TCP traffic

Adaptive Coding Schemes (adaptive Forward Error Control – FEC)

- CS 1: 9.05 Kbps/slot
- CS 2: 13.4 Kbps/slot
- CS 3: 15.6 Kbps/slot
- CS 4: 21.4 Kbps/slot (no FEC)



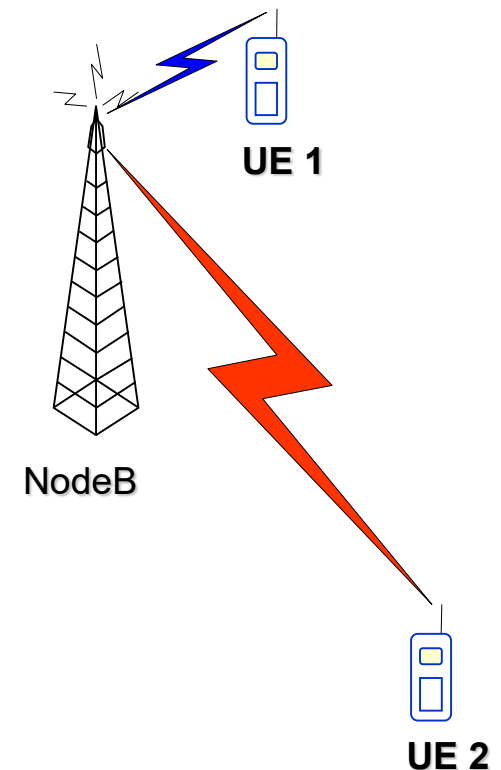
Problems and limits

- IP-based network => high latency, no guarantees
- Limited data rate: 28 kbps (3 slot/CS-1) - 64.2 kbps (3 slot/CS-4)
- Latency/flow control problems with TCP

EDGE (Enhanced Data Rates for GSM Evolution)

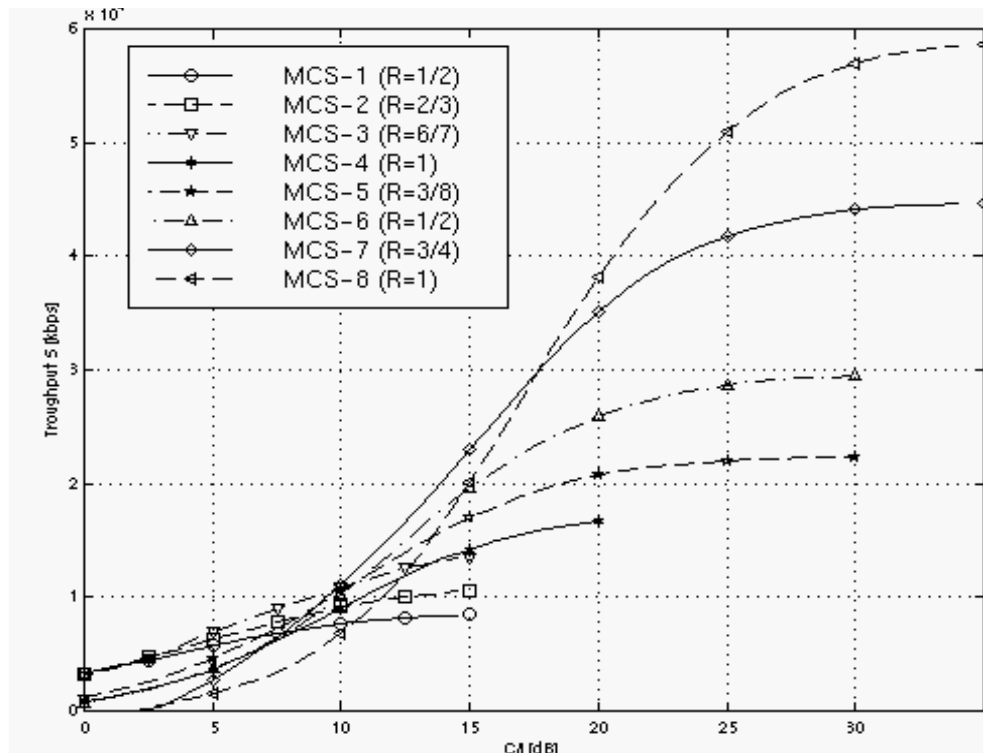
- EDGE can carry data speeds up to 236.8 kbit/s for 4 timeslots
- theoretical maximum is 473.6 kbit/s for 8 timeslots
- **Adaptation of modulation** depending on quality of radio path
 - GMSK (GSM standard – 1 bit per symbol)
 - 8-PSK (3 bits per symbol)
- **Adaptation of coding scheme** depending on quality of radio path (9 coding schemes)
- Gain: data rate (gross) up to 69,2kbps (compare to 22.8kbps for GSM)
- EDGE is a complex extension of GSM!

Near-far problem

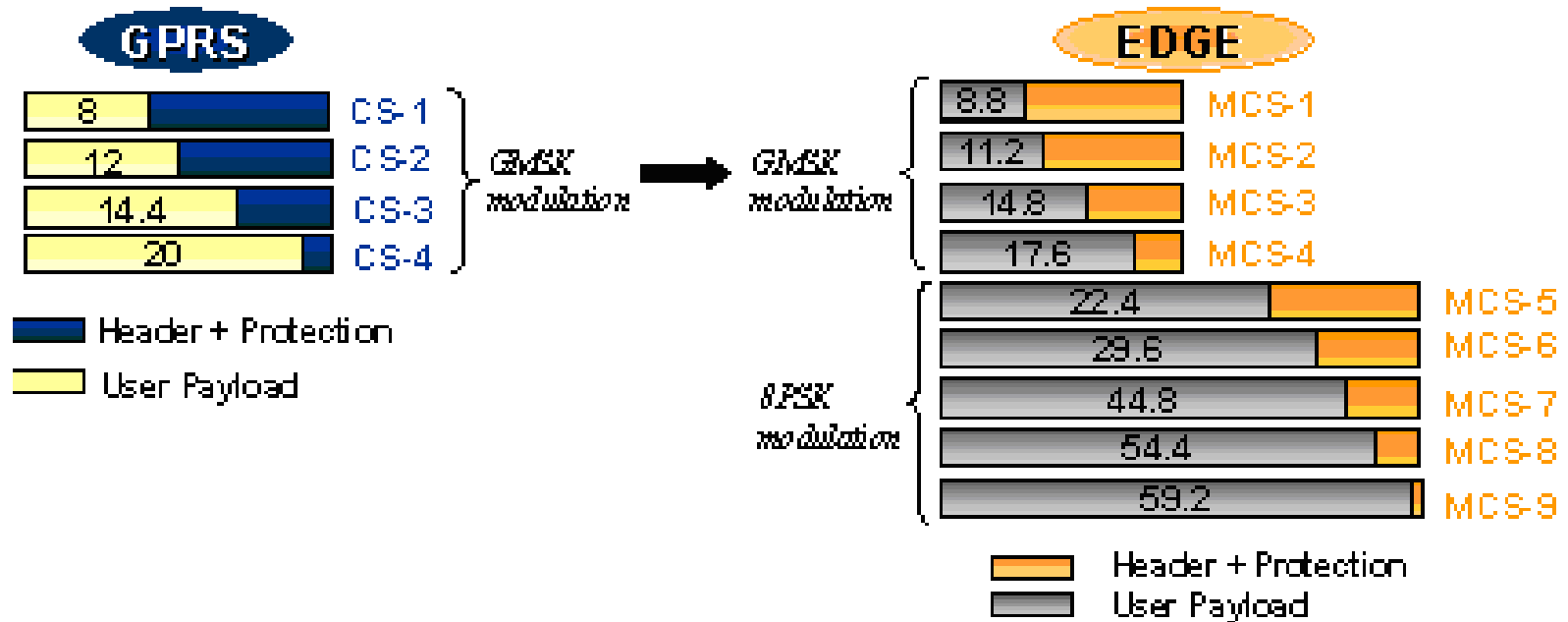


EDGE – Adaptive Modulation and Coding Schemes

Scheme	Modulation	Maximum rate [kb/s]	Code Rate	Family
MCS-9	8PSK	59.2	1.0	A
MCS-8		54.4	0.92	A
MCS-7		44.8	0.76	B
MCS-6		29.6 / 27.2	0.49	A
MCS-5		22.4	0.37	B
MCS-4	GMSK	17.6	1.0	C
MCS-3		14.8 / 13.6	0.80	A
MCS-2		11.2	0.66	B
MCS-1		8.8	0.53	C



Payload for GPRS and EDGE



Differences of GSM/GPRS Compared to WLAN Systems

Spectrum management and utilization

- coverage and interference management due to cell planning
- capacity and load management due to admission control and QoS support
- flexible cell size simplifies cost-efficient nationwide coverage

Mobility management

- fast, lossless HO due to make-before-break

General control structures and control philosophy

- high reliability and QoS guarantees due to centralized/infrastructure-based management and control of all resources

Energy

- high energy cost on network side, low cost on mobile due to passive cell camping instead of active association, paging mode and sleep cycles

Customer relations

- monthly/bi-yearly contracts, pay per service
- security due to preshared credentials

Implementation

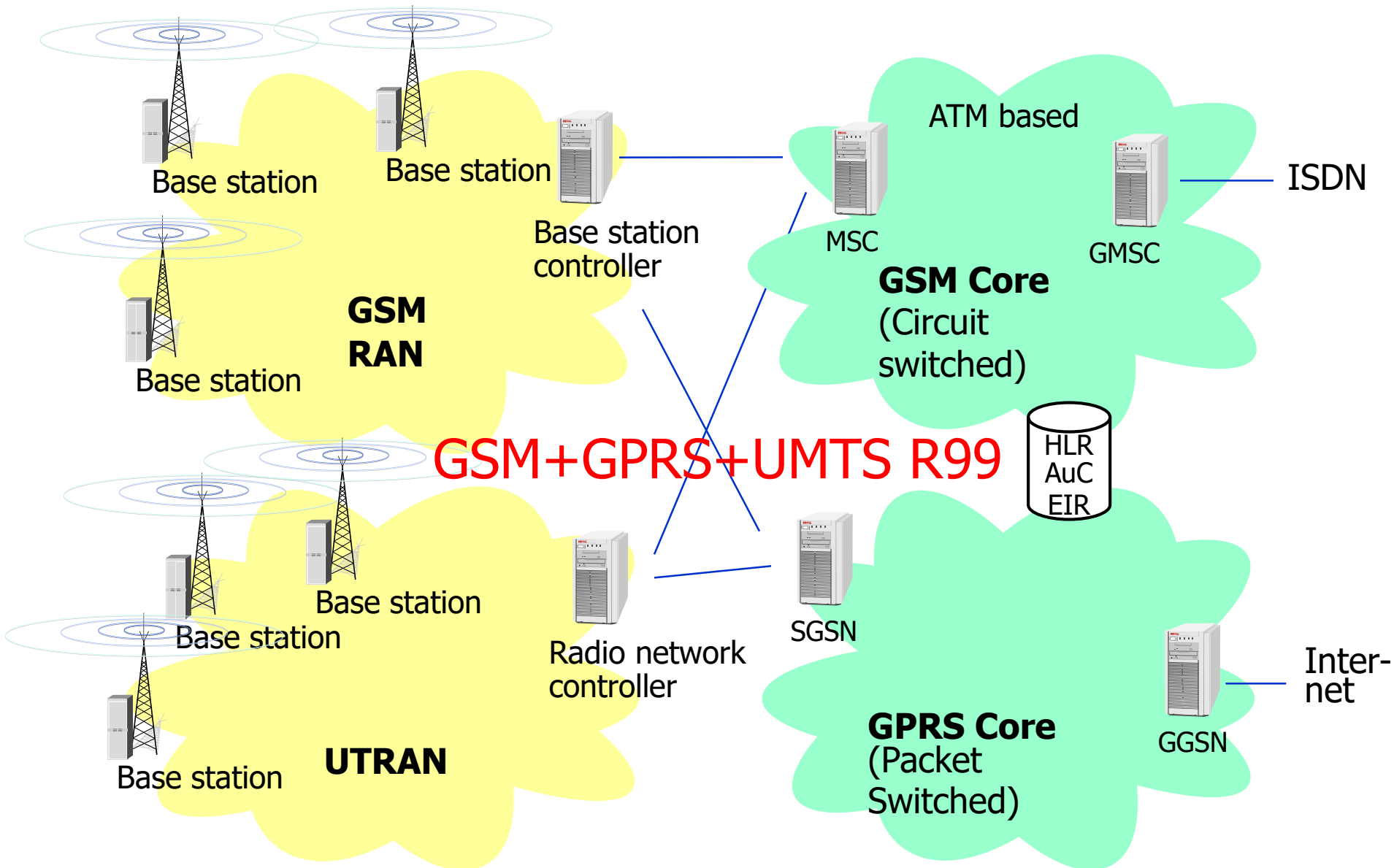
- simple implementation of TDMA, e.g. with GNUradio and SDR

UMTS (Universal Mobile Telecommunication System)

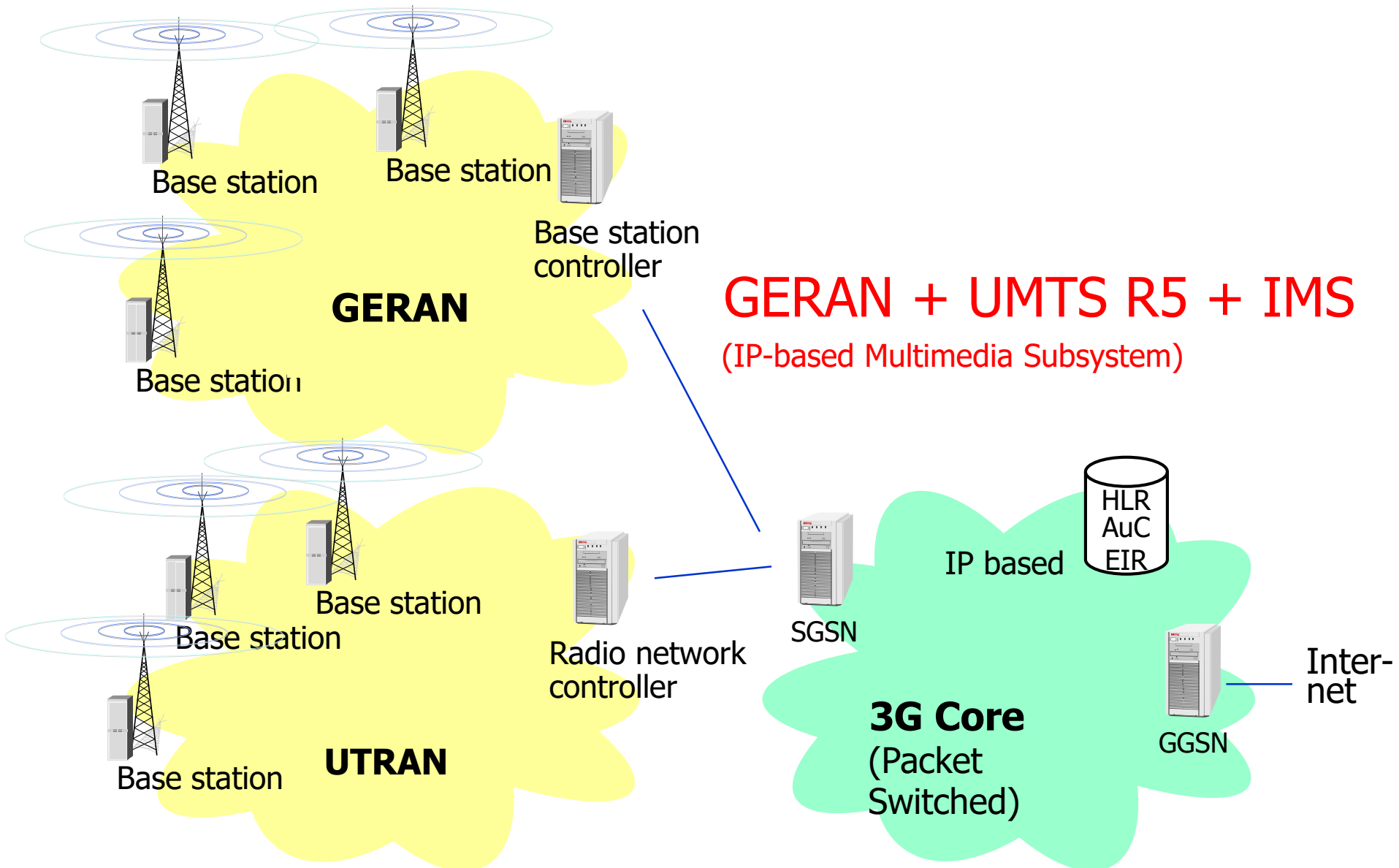
Questions to be answered:

- Which multiplexing scheme is employed by UMTS?
- What does macro diversity mean?
- What does cell breathing in UMTS mean?
- How does mobility management work in UMTS?
- How does QoS management work in UMTS?

2G to 3G Evolution: GSM - GPRS – UMTS R99



2G to 3G Evolution: GSM - GPRS - UMTS R5 - IMS



UMTS – Macro Diversity in CDMA

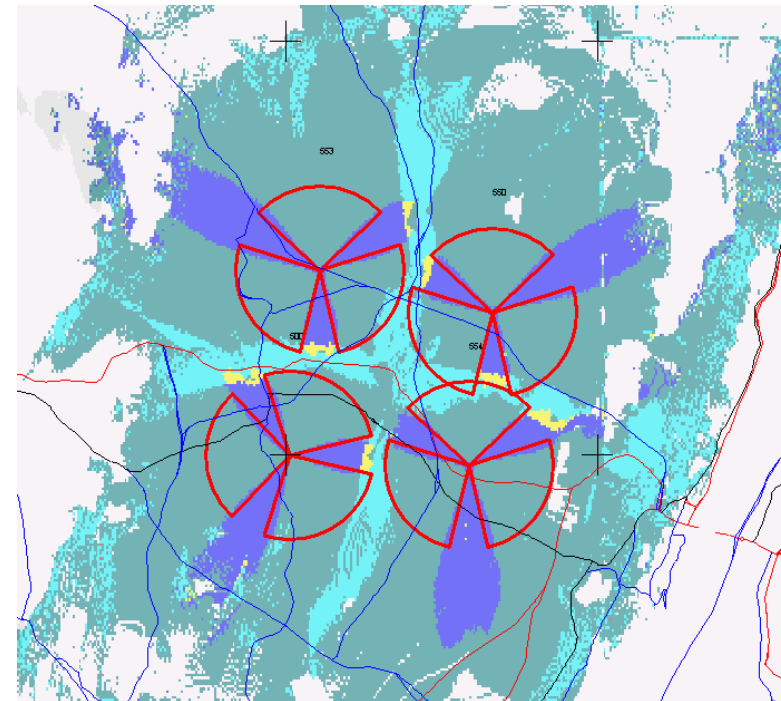
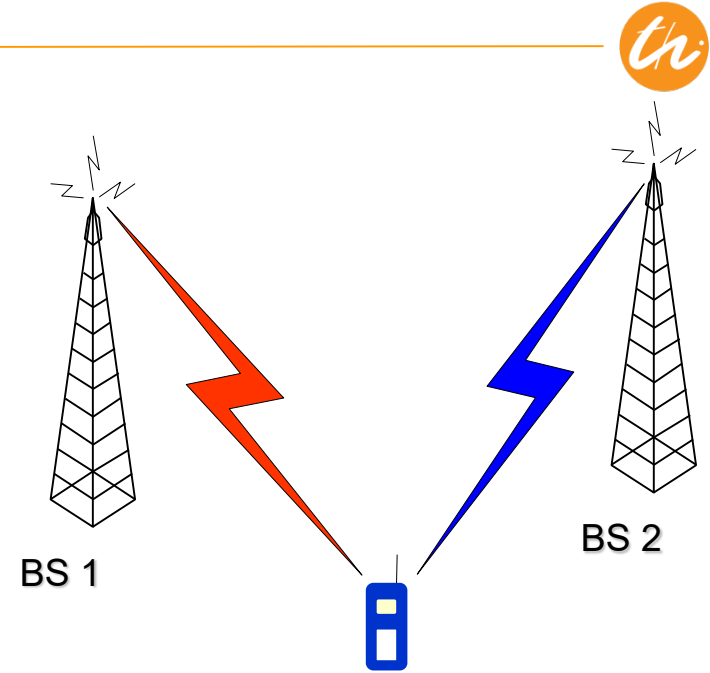
CDMA is a single frequency system

Mobile user may be served

- by multiple base stations
 - => soft handover
 - => combining in RNC
- by multiple cells from the same base station
 - => softer handover
 - => combining in base station

Due to increased redundancy, the combining of signals improves quality (BER) or allows to reduce required TX power to achieve given BER requirements.

Note that this is an extension of the make-before-break strategy where radio links are continuously added and deleted.

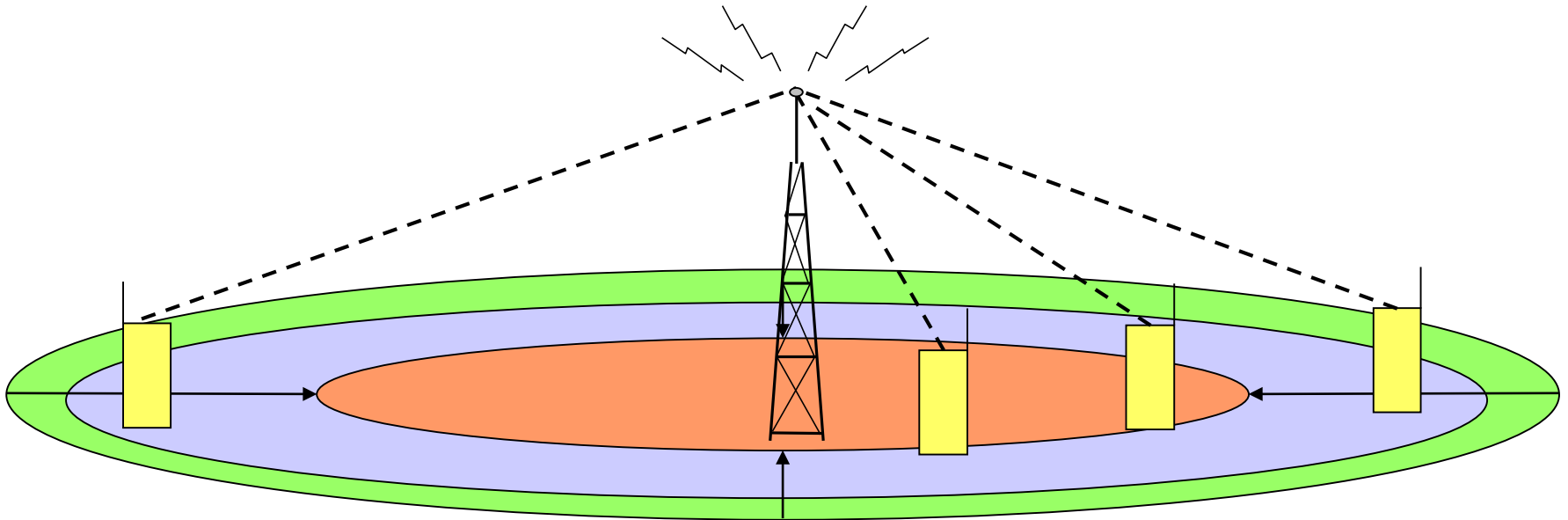


UMTS – Cell Breathing

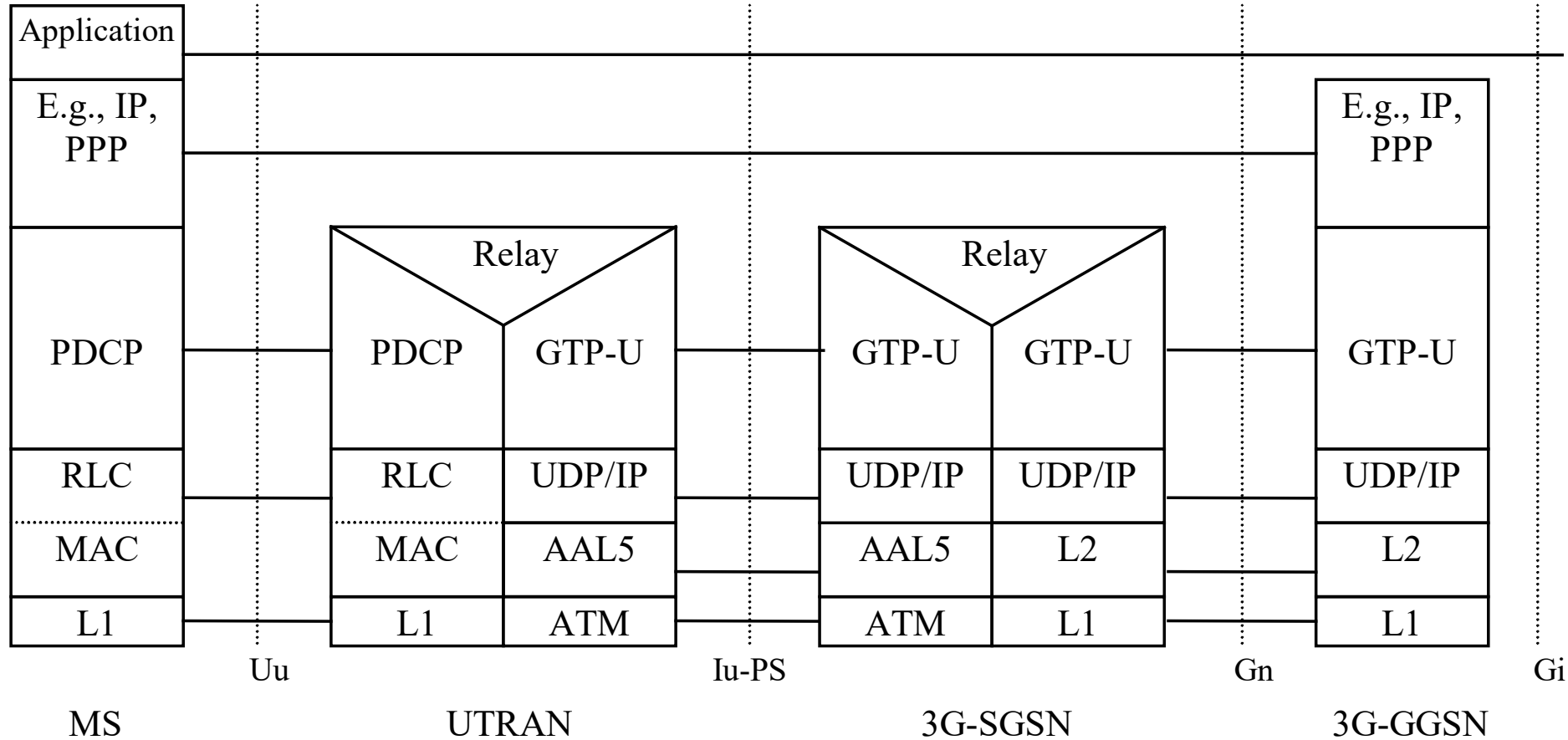
Issue in CDMA systems: cell size depends on the actual load

Effects:

- Additional traffic (i.e. more CDMA codes used) will cause higher interference to users of the same cell
 - If the interference becomes too strong, users at the cell edge can no longer meet SINR requirements to continue communication
- ⇒ Effective coverage area of the cell reduces, i.e. cell shrinks

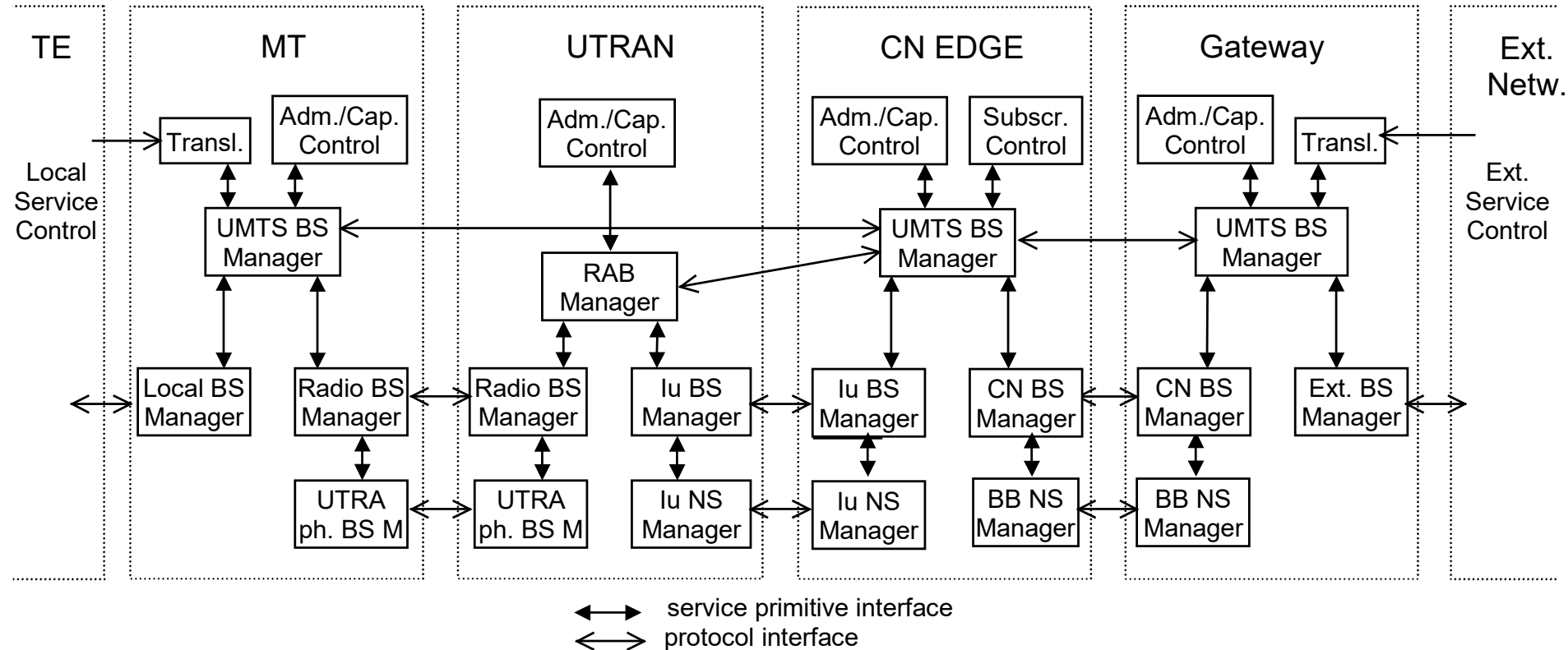


UMTS – Mobility Management in Packet Domain



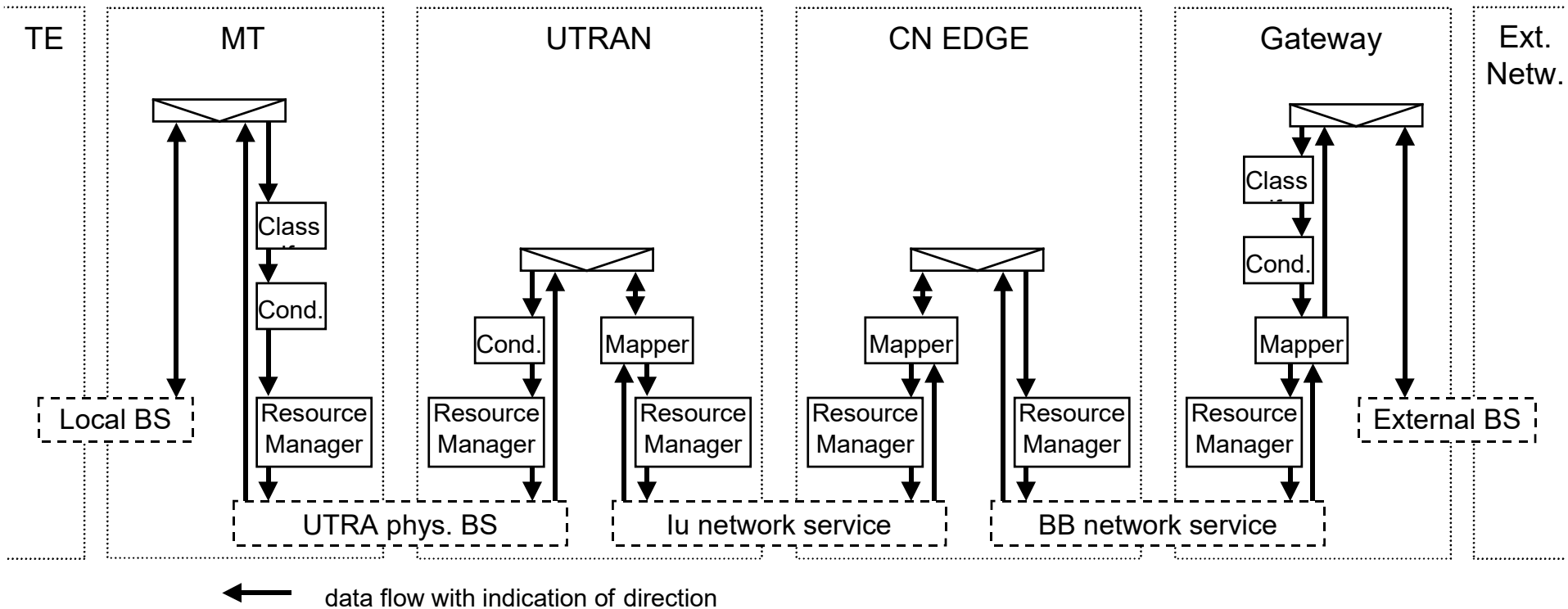
End-to-End Resource Management in UMTS (contr. plane)

A sophisticated QoS architecture



For details see 3GPP TS 23.207

End-to-End Resource Management in UMTS (user plane)



LTE (Long-Term Evolution)

Questions to be answered:

- What is the advantage of a common IP-based packet core over separated networks for packet-switched and circuit-switched traffic?
- What is the purpose of opportunistic scheduling? What is the disadvantage?
- How does interference cancelation work? What is the idea?
- What is the goal of self-organization in LTE?
- What are the use cases for self-organized network management in LTE?

Evolution from GSM to UMTS and LTE

GSM: voice-dominated, dedicated channels, heavy states

GPRS: adds support for packet data on shared channels; adds IP-based core network

EDGE: increased packet data capacity employing adaptive modulation

UMTS: separate voice and packet data support; focus on dedicated channels and heavy states, complicated RAN architecture and protocols due to macro diversity (soft handover) and QoS requirements

HSPA: improved support for packet data; emphasis on shared channels and fast radio resource management

IMS: support for IP-based multimedia services, e.g. voice (VoIP) or video

LTE: strong packet data support (latency, throughput, control overhead), limited state; simplified protocols; PS only, i.e. no CS core network

Evolution of technology

- from circuit switching to packet switching
- from slow, explicit setup and release of resources to fast channel-condition- and demand-specific resource management

Evolution towards LTE – Architecture

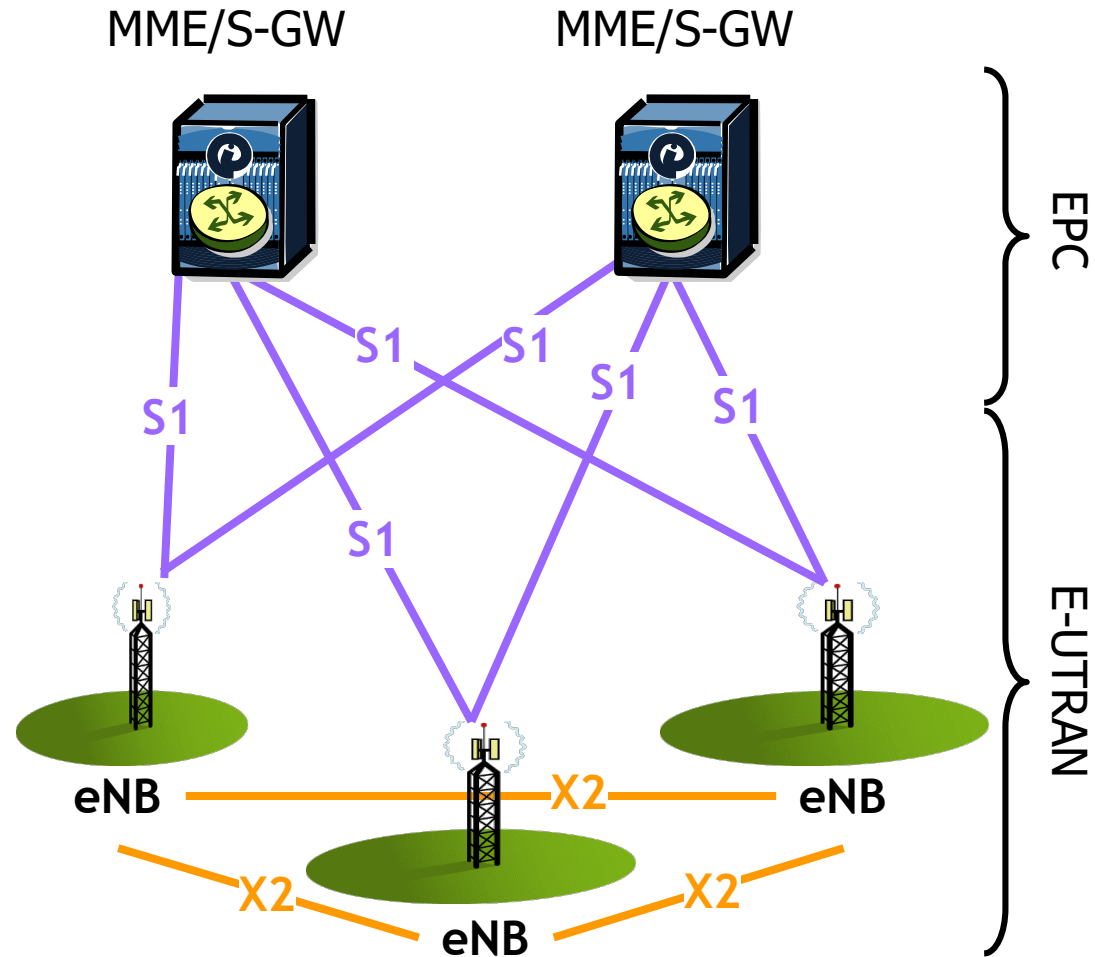
- LTE radio system is a **packet-only network** – there is no support for circuit-switched services (no MSC)
- LTE started on a **clean state** - everything was up for discussion including the system architecture and the split of functionality between Radio Access Network (RAN) and Core Network (CN)
- 3GPP (3rd Generation Partnership Program) study items
 - „**3G Long-term Evolution**” (LTE) for new Radio Access and
 - “**System Architecture Evolution**” (SAE) for Evolved Network

LTE: Evolved Packet System (EPS) Architecture

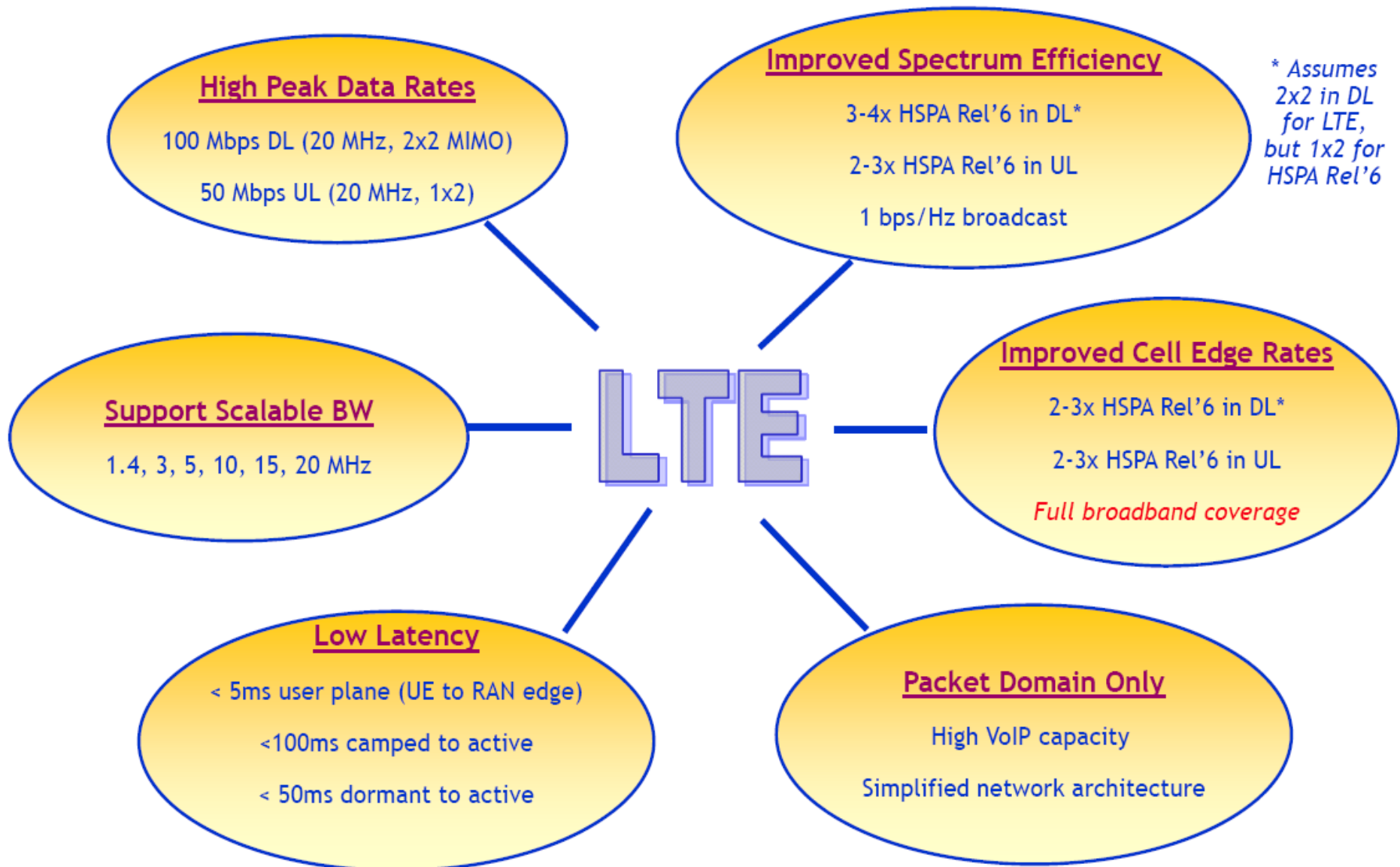
2 instead of 4 user plane entities

Key changes

- eNB: merging of base station and RNC functionality
- S-GW: merger of SGSN and GGSN functionality



LTE: Requirements and Performance Targets



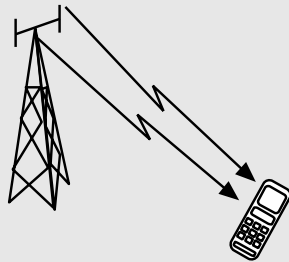
LTE Key Features (Release 8)

- Support for both FDD and TDD
- Adaptive modulation and coding
 - DL modulations: QPSK, 16QAM, and 64QAM
 - UL modulations: QPSK and 16QAM
- Hybrid ARQ in addition to ARQ
- Multi-antenna solutions
 - (2 or 4) x (2 or 4) downlink and uplink supported
 - Multi-layer transmission with up to four streams
 - Multi-user MIMO also supported
- Support for interference coordination

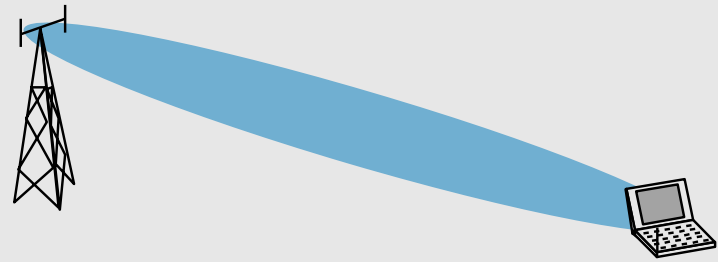
Multi-antenna Solutions

Different antenna solutions needed for different scenarios/targets

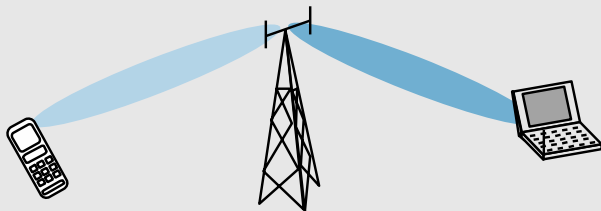
- *High peak data rates* ➔ *Multi-layer transmission*
- *Good coverage* ➔ *Beam-forming*
- *High capacity* ➔ *Beam forming (and multi-layer transmission)*



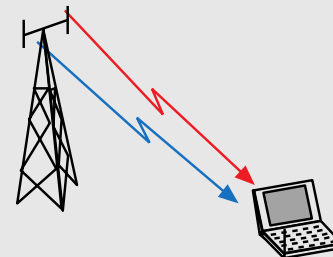
Diversity for improved system performance



Beam-forming for improved coverage
(less cells to cover a given area)



Spatial-division multiple access
("MU-MIMO") for improved capacity
(more users per cell)

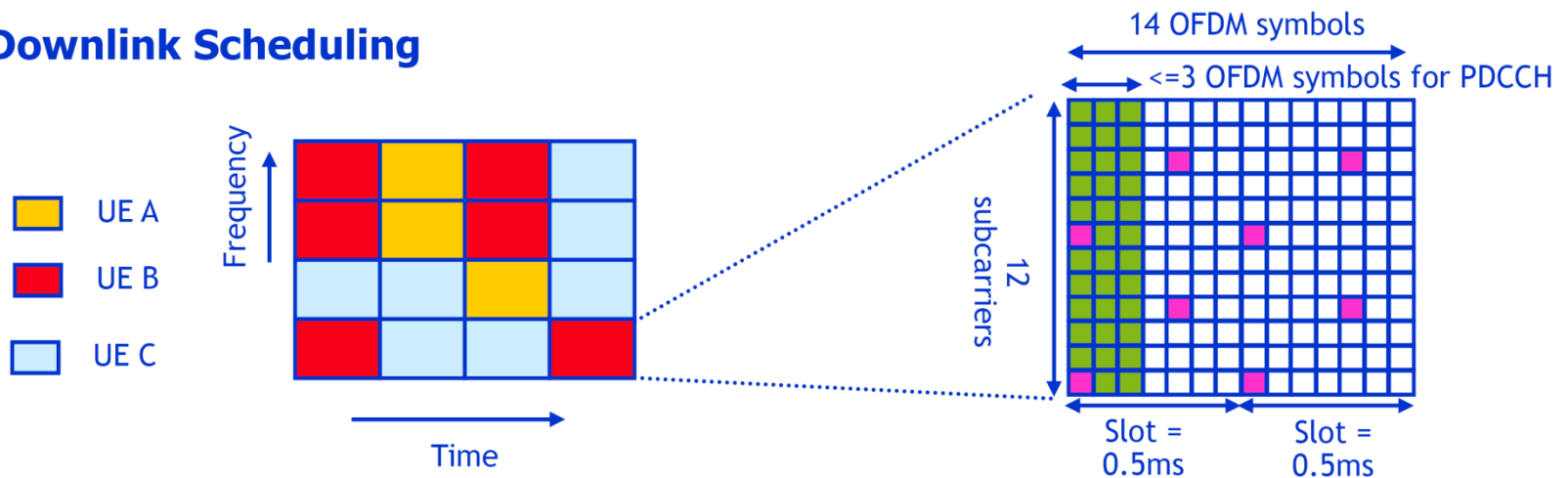


Multi-layer transmission
("SU-MIMO") for higher data rates
in a given bandwidth

Scheduling and Resource Allocation

- Fast scheduling
- Scheduled, shared channel on both uplink and downlink
=> all transmissions in UL and DL must be explicitly scheduled

Downlink Scheduling

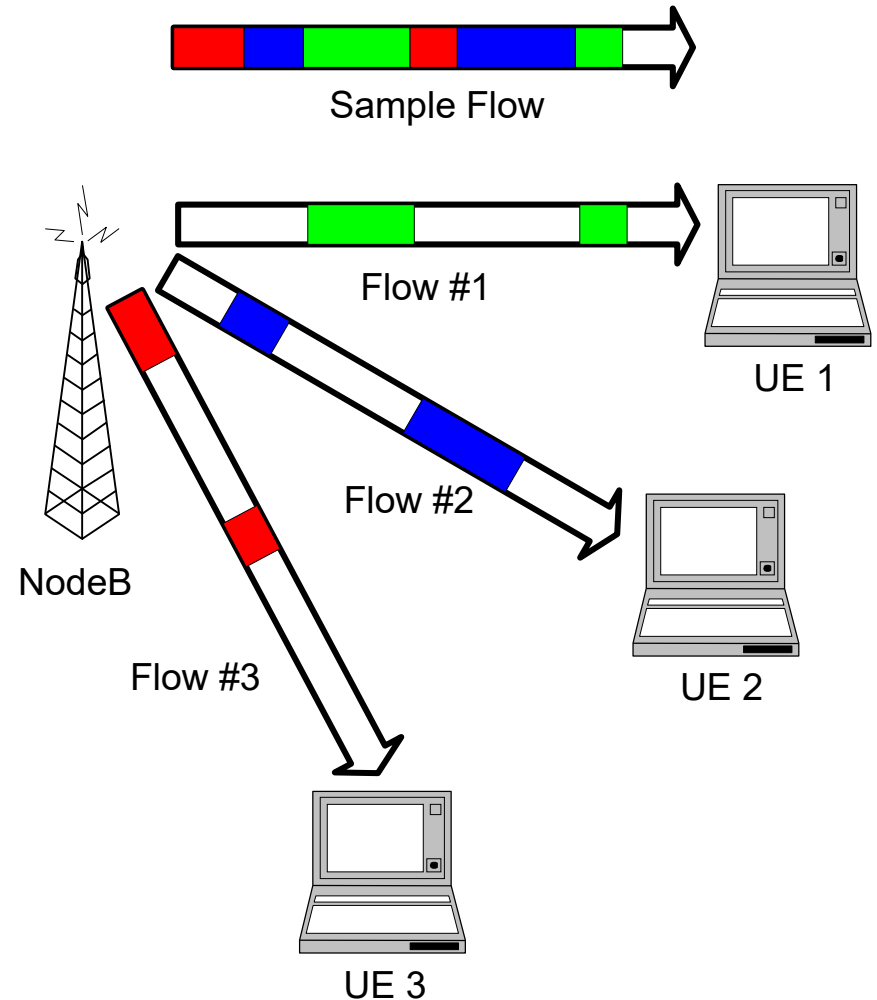


- Support for "semi-persistent" (periodical) allocation of resources for low-latency traffic, e.g. VoIP

Opportunistic Scheduling

Prefer mobiles with good channel quality
=> maximizes system throughput

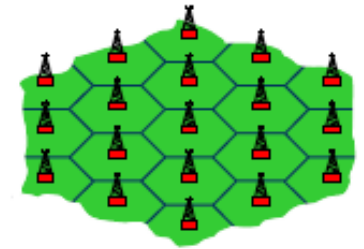
Ensure fairness for users with
continuously poor channel quality



Interference Coordination

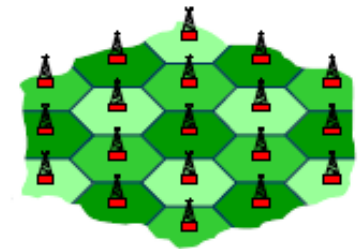
High data rates in limited spectrum allocations

- Entire spectrum must be available in each cell
- ➔ One-cell frequency reuse



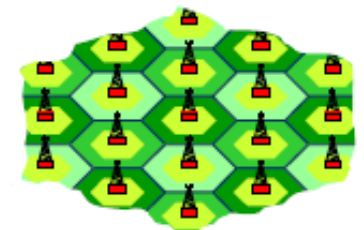
Reduced inter-cell interference with frequency reuse > 1

- Improved cell-edge SIR ➔ Higher cell-edge data rates



Adaptive reuse

- Cell-center users: Reuse = 1
- Cell-edge users: Reuse > 1



Self-Organization in LTE

Goal: minimize OPEX by automation of planning, optimization and repair of the network

Use Cases for Self-organized Network Management

- Physical cell-ID automatic configuration (PCI)
- Automatic Neighbor Relation (ANR)
- Coverage and capacity optimization (CCO)
- Inter-cell interference coordination (ICIC)
- Random Access Channel (RACH) optimization
- Mobility load balancing optimization (MLB)
- Mobility robust optimization (MRO)
- Energy saving (power on/off)

References

LTE/SAE

- A. Toskala et al, "UTRAN Long-Term Evolution," Chapter 16 in Holma/ Toskala: WCDMA for UMTS, Wiley 2007
- E. Dahlman et al, "3G Evolution, HSPA and LTE for Mobile Broadband," Elsevier Journal, 2007
- Special Issue on LTE/ WIMAX, Nachrichtentechnische Zeitung, pp. 12–24, 1/2007
- 3rd Generation Partnership Project Long Term Evolution (LTE), official website:
<http://www.3gpp.org/Highlights/LTE/LTE.htm>
- Technical Paper, "UTRA-UTRAN Long Term Evolution (LTE) and 3GPP System Architecture Evolution (SAE)", last update October 2006, available at:
ftp://ftp.3gpp.org/Inbox/2008_web_files/LTA_Paper.pdf

Standards

- TS 36.xxx series, RAN Aspects
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